Fourth-Order Perturbations of Pure-Weyl Black Holes

Exact Ricci–Bach Composition, Horizon Pairing, and Endpoint Nonselection

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Abstract

We ask whether familiar one-ended Lorentzian endpoint conditions isolate the Einstein sector of four-dimensional pure Weyl gravity around a black hole. We first classify the Bach-flat locus in a declared static Laurent ansatz, recover the Mannheim–Kazanas component, and derive a residual-basic horizon generator. On its nondegenerate static domain, the associated Lee–Wald Hamiltonian, Wald entropy, and signed surface-gravity temperature obey an exact first law at every simple horizon. We correct an important sheet distinction: on the complete Laurent locus the metric is Einstein precisely when $\gamma = 0$ and $w = 1$, whereas $\gamma = 0$ alone suffices only on the Mannheim–Kazanas component through $w = 1$.

On Schwarzschild, linearizing the Bach tensor gives an exact Ricci-carrier composition. It defines an exact sequence from Bach solutions to curvature solutions, not a canonical Einstein/additional direct sum. In the axial $\ell = 2$ carrier, $\psi_{ab} = \delta R_{ab}$ obeys a second-order Lichnerowicz-type equation. For nonzero real frequency its horizon system has a two-dimensional analytic ingoing ψ-space, so future-horizon regularity does not force $\psi = 0$. The action-derived Lee–Wald current makes the Einstein self-block isotropic for conjugate Regge–Wheeler waves. Controlled finite-series fixtures at three frequencies exhibit nonzero Einstein–additional cross pairing and nonzero additional-sector pairing; these fixture results are not promoted to a symbolic all-frequency theorem.

At infinity the earlier $\ell = 2$, $M_{\mathrm{BH}}\omega = 3/5$ claim that finite radial pairing selects exactly the Einstein sector is superseded. A Phase-3 audit found that the axial metric reconstruction had also omitted the independent $v\varphi$ Ricci equation. On the declared $\ell = 2$, $M_{\mathrm{BH}} = 1$, $\omega \in [1/2, 3/4]$ pilot this equation propagates as $C' = -2C/r$; its zero fibre is an exact six-dimensional formal module. The repaired non-Einstein X_0 direction is radially finite in the fixed representative and under the true rate-zero Einstein shear, but its cross pairing with the true oscillatory Einstein shear diverges. Thus no representative-independent axial selection theorem follows. Independently, the polar restriction-stable module contains a mixed Einstein/additional line that is finite and generically nonradical, with an explicit algebraic exceptional locus. Thus neither leading endpoint diagnostics nor finite formal radial pairing isolates the Einstein image. These are formal infinity-mode statements: no horizon-to-infinity matching, asymptotically flat phase space, scattering, stability, quasinormal spectrum, Hawking state, or quantum unitarity theorem is claimed. The local Cauchy restriction $\psi|_\Sigma = \nabla_n\psi|_\Sigma = 0$ still selects the Einstein kernel at the linear level.

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AI authorship and computational accountability

The named AI systems developed the programme, derivations, symbolic implementations, independent verification architecture, claim audits, and this manuscript. Asger Alstrup Palm commissioned and coordinated the work and is the corresponding human contact. Every certified statement below is bound to a machine-readable certificate and a separately implemented verifier. The cross-flux result uses controlled high-precision series evaluation with exact recurrence data and independent frequencies; it is reported as fixture-level evidence, not interval-certified nonvanishing. All local tensor identities and polynomial classifications are exact. The manuscript remains conditional on final human review.

Contents

1 The physical question

Fourth-order gravitational equations contain more local solutions than the Einstein equations. A common proposal is to retain the Einstein branch by a boundary condition. Around a one-ended Lorentzian black hole, a first test is:

> Do future-horizon analyticity and the same leading outer admissibility tests used for Einstein waves force the additional Ricci carrier to vanish?

The answer in the certified Schwarzschild axial $\ell = 2$ mode laboratory is no: the curvature carrier has regular ingoing horizon data, the leading real-frequency symbol does not separate it from the Einstein carrier, and controlled fixtures have nonzero action-derived horizon pairing. This is an endpoint-admissibility result, not a classification of all local differential boundary conditions. In particular, setting the Ricci carrier and its normal derivative to zero on a Cauchy surface is a local linear Einstein-sector restriction. Whether that restriction is natural, interaction-stable, or compatible with a physical asymptotic phase space is open.

1.1 Claims and dependency tags

The static family and local operator identities carry LOCAL-ALGEBRAIC. Harmonic branch, horizon, current, and asymptotic statements also carry REDUCED-MODE. No result in this manuscript is promoted to LORENTZIAN-CAUSAL: there is no complete exterior Green-hyperbolicity theorem, metric-falloff phase space, or classified set of local differential boundary operators.

2 Relation to existing branch-selection programmes

The static Bach-flat family itself is classical [2, 3]. Conformal-gravity thermodynamics with additional spin-two hair has also been studied in asymptotically AdS and Einstein–Weyl settings [14]. Our static contribution is therefore not a new family: it is the exact residual-basic normalization and simultaneous signed first-law identity within a declared local quotient. A physical asymptotic normalization has not yet been selected. In particular, Lü, Pang, Pope, and Vázquez-Poritz organize AdS thermodynamics with an independent massive-spin-two-hair conjugate pair. Our one-term identity instead follows after quotient-basic normalization on a restricted static parameter family. The different boundary ensemble, generator normalization, and independent variables mean that the two first laws are complementary, not contradictory.

Maldacena's Einstein-from-conformal construction selects an Einstein sector by Euclidean AdS or late-time de Sitter boundary data [8]. Critical- and logarithmic-gravity analyses similarly separate massless, massive, and generalized modes by asymptotic conditions [9, 13, 10, 11, 12]. The null Einstein self-pairing and nonzero cross pairing found below should be read in that established degenerate-operator context, rather than as the first appearance of null massless modes paired with generalized solutions. Our different question is whether one-ended Lorentzian Schwarzschild endpoint tests force the Ricci carrier to vanish.

Using δR_{ab} to expose higher-derivative spin-two modes is also a known strategy in fourth-order gravity [15]. Previous Weyl-black-hole quasinormal analyses often deliberately obtained a second-order master equation by conformal transport rather than retaining the full fourth-order Bach solution space [16]. Effective-Einstein perturbations of conformally related nonsingular metrics [17] and scalar or electromagnetic test-field spectra [18] are different problems: test-field stability is not stability of the dynamical fourth-order Bach sector. Contemporary quadratic-gravity ringdown work finds both ordinary and additional spin-two sectors [19], while time-domain quadratic-gravity studies track massive-sector tails [20]. The prospective advance here is narrower: the exact pure-Weyl Ricci–Bach sequence, its action-derived Schwarzschild horizon pairing, and failure of the tested endpoint conditions to imply $\delta R_{ab} = 0$. The first-order-system analysis follows the general modified-gravity asymptotic methodology of [21]; in particular, that methodology makes the unresolved repeated-root Jordan audit an essential next step rather than a technical detail. Precise decay of the metric coefficients must ultimately be compared with the Schwarzschild Einstein benchmark [22], not inferred from bounded curvature carrier variables alone.

3 Pure Weyl gravity and its covariant current

We use signature $(-+++)$ and action

$$S_W[g] = \alpha \int_M \sqrt{-g}\, C_{abcd} C^{abcd}\, \mathrm{d}^4 x. \tag{1}$$

The field equation is the Bach equation $B_{ab} = 0$, with convention

$$B_{ab} = \nabla^c \nabla^d C_{acbd} + \frac{1}{2} R^{cd} C_{acbd}. \tag{2}$$

The covariant presymplectic current is obtained from the literal variation of the curvature-momentum potential of this action. Its normalization is fixed by the off-shell Green identity

$$\partial_t F^t(h_A, h_B) + \partial_r F^r(h_A, h_B) = 4\alpha \int_{S^2} \sqrt{q}\,\left(h_B^{ab} \delta B_{ab}[h_A] - h_A^{ab} \delta B_{ab}[h_B]\right). \tag{3}$$

Thus the current is conserved for pairs of linearized Bach solutions and is not a freely rescaled master-equation Wronskian.

4 Static spherical Bach-flat family

In Schwarzschild gauge,

$$\mathrm{d}s^2 = -B(r)\mathrm{d}t^2 + B(r)^{-1}\mathrm{d}r^2 + r^2 \mathrm{d}\Omega_2^2, \tag{4}$$

the declared Laurent ansatz is

$$B(r) = w - \frac{u}{r} + \gamma r - k r^2 + \frac{c_2}{r^2} + c_3 r^3. \tag{5}$$

Theorem 4.1 (Complete Bach-flat locus in the Laurent ansatz)**.** *The metric is Bach-flat if and only if*

$$c_2 = c_3 = 0, \qquad w^2 + 3u\gamma = 1. \tag{6}$$

The Mannheim–Kazanas parametrization is recovered by

$$w = 1 - 3\beta\gamma, \qquad u = \beta(2 - 3\beta\gamma).$$

On the complete Laurent locus the metric is Einstein if and only if

$$\gamma = 0, \qquad w = 1. \tag{7}$$

On the Mannheim–Kazanas component through $w = 1$, this reduces to the single condition $\gamma = 0$.

Completeness here refers only to the Laurent ansatz. The certificate does not claim a new proof of the global static spherical classification. The sheet qualification in (7) follows already from the scalar curvature

$$R = 12k - \frac{6\gamma}{r} + \frac{2(1-w)}{r^2}. \tag{8}$$

In particular, the allowed point $(\gamma, w) = (0, -1)$ is not Einstein. The original static certificate's phrase "Einstein iff $\gamma = 0$" is therefore used only on its explicitly parametrized Mannheim–Kazanas sheet; the manuscript does not propagate it to the complementary Laurent sheet.

The form-preserving local Diff $\times$Weyl transformations act as dilation and translation on the cubic

$$Q(x) = -ux^3 + wx^2 + \gamma x - k, \qquad x = r^{-1}.$$

Their orbit rank is two. A single continuous invariant may be taken as

$$J = u^2 \operatorname{disc}(Q),$$

while $\operatorname{disc}(Q) = 0$ is the degenerate-horizon locus. The conformal translation may fail globally where its Weyl factor vanishes; it is used only as a local orbit statement.

An exact non-Einstein three-horizon fixture is

$$(\beta, \gamma, k) = \left(\frac{3}{2}, \frac{12}{19}, \frac{1}{19}\right), \qquad rB(r) = -\frac{1}{19}(r-1)(r-3)(r-8).$$

All curvature invariants are finite at the simple horizons. In ingoing Eddington–Finkelstein coordinates,

$$\mathrm{d}s^2 = -B(r)\mathrm{d}v^2 + 2\mathrm{d}v\mathrm{d}r + r^2\mathrm{d}\Omega_2^2,$$

the metric and Bach equation are regular there.

5 Residual-basic energy and the static first law

For the chart generator ∂_t, the bare static Lee–Wald one-form is not integrable. The obstruction disappears when the generator is normalized consistently with the residual quotient.

Theorem 5.1 (Normalized static generator)**.** *On a connected component with $u \neq 0$, choose the oriented generator*

$$\chi = u\,\partial_t, \qquad u = \beta(2 - 3\beta\gamma). \tag{9}$$

Residual basicness and dilation weight determine the general normalization as $uf(J)\partial_t$. We select $f(J) = 1$. Its Hamiltonian is

$$H = -16\pi\alpha\beta^2 D_2, \qquad D_2 = 9\beta^2\gamma^2 k - \beta\gamma^3 - 12\beta\gamma k + \gamma^2 + 4k. \tag{10}$$

At every simple horizon r_h, the Wald entropy

$$S_h = 64\pi^2\alpha\beta\frac{2 - 3\beta\gamma + \gamma r_h}{r_h} \tag{11}$$

and normalized temperature

$$T_h = \frac{uB'(r_h)}{4\pi} \tag{12}$$

satisfy

$$\mathrm{d}H = T_h\,\mathrm{d}S_h \tag{13}$$

identically modulo $B(r_h) = 0$.

Here T_h is the *signed* surface-gravity temperature appropriate to the oriented first-law identity; the positive Hawking temperature would use $|uB'(r_h)|/(4\pi)$. On a component where $u < 0$, future-directedness uses $-\chi$, flipping both H and T_h and preserving the first law. The locus $u = 0$ is excluded because this normalization degenerates. For another allowed $f(J)$, one has $T_f = f(J)T_h$ and a reparametrized Hamiltonian satisfying $\mathrm{d}H_f = f(J)\mathrm{d}H$; the first law remains true. Thus residual basicness fixes a class, not a unique physical normalization in the absence of an asymptotic clock.

This is a theorem on the static parameter slice, not yet a general physical-process first law. Its linear dynamical extension nevertheless passes a strong gauge audit: arbitrary time-dependent spherical Weyl rescalings and diffeomorphisms have exactly zero charge and flux, the entropy is conformally invariant on the symbolic family, and the normalized generator extension is unique at linear order under the declared boundary scalar rule.

6 Ricci–Bach composition and the axial exact sequence

The parity-specific formulas are restrictions of one Ricci-flat identity.

Proposition 6.1 (Linearized Bach tensor on a Ricci-flat background)**.** *For a perturbation of a four-dimensional Ricci-flat metric,*

$$\delta B_{ab} = \frac{1}{2}\Box(\delta R_{ab}) + C_{acbd}(\delta R)^{cd} - \frac{1}{6}\nabla_a\nabla_b\delta R - \frac{1}{12}g_{ab}\Box\delta R. \tag{14}$$

Proof. Rewrite the four-dimensional Bach tensor in Ricci variables using the contracted Bianchi identity, then retain the terms linear in Ricci curvature about $R_{ab} = 0$. Background derivatives and the curvature commutator give the displayed Weyl action. The repository verifies (14) independently, component by component, on the complete axial and polar $\ell = 2$ Regge–Wheeler-gauge carriers. That computer check is carrier-complete but is not presented as a machine proof for arbitrary harmonic content. □

We now specialize the background to Schwarzschild and use axial Regge–Wheeler gauge,

$$h_{t\phi} = h_0(t,r)S_{\ell m}, \qquad h_{r\phi} = h_1(t,r)S_{\ell m}.$$

The $\ell = 2$ linearized Bach tensor is fourth order, trace-free, and obeys the exact linearized Noether identity. Eliminating the constraint on the Einstein subbranch reproduces the Regge–Wheeler equation for $\Psi = Bh_1/r$,

$$B(B\Psi')' + (\omega^2 - V_{\mathrm{RW}})\Psi = 0, \qquad V_{\mathrm{RW}} = B\left(\frac{6}{r^2} - \frac{6m}{r^3}\right). \tag{15}$$

Corollary 6.2 (Axial composition)**.** *For the certified axial carrier, parity gives $\delta R = 0$, so*

$$\boxed{\delta B_{ab} = \frac{1}{2}\Box(\delta R_{ab}) + C_{acbd}(\delta R)^{cd}.} \tag{16}$$

Consequently the Einstein kernel $\delta R_{ab} = 0$ injects into the Bach kernel, while its curvature image obeys

$$\frac{1}{2}\Box\psi_{ab} + C_{acbd}\psi^{cd} = 0, \qquad \psi_{ab} := \delta R_{ab}. \tag{17}$$

Let $\mathcal{C}_{\mathrm{ax}}$ be the fixed axial carrier and define

$$\mathcal{H}_B = \{h \in \mathcal{C}_{\mathrm{ax}} : \delta B[h] = 0\}, \qquad \mathcal{H}_E = \{h \in \mathcal{H}_B : \delta R[h] = 0\}, \tag{18}$$

$$\mathcal{H}_R = \ker\big(\tfrac{1}{2}\Box + C\circ\big) \cap \mathrm{im}(\delta R|_{\mathcal{C}_{\mathrm{ax}}}). \tag{19}$$

Then the invariant statement supplied by the composition is the exact sequence

$$0 \longrightarrow \mathcal{H}_E \longrightarrow \mathcal{H}_B \xrightarrow{\delta R} \mathcal{H}_R \longrightarrow 0. \tag{20}$$

It does *not* supply a canonical direct sum. A metric lift h_X of $\psi \in \mathcal{H}_R$, when it exists in the chosen domain, may be shifted by any $h_E \in \mathcal{H}_E$. Global lift existence, boundary conditions, and a gauge-independent splitting must therefore be proved separately. In particular, "additional branch" below refers either to the curvature image $\mathcal{H}_R$ or to an explicitly constructed lift, never to an assumed canonical complement.

The identity is componentwise and off shell in the perturbation. It does not extend in this two-term form to the certified non-Einstein static fixture. Thus the clean composition is a Ricci-flat-background theorem.

7 The Ricci carrier reaches the future horizon

In ingoing coordinates (v, r), Fourier modes are written $e^{i\omega v}$. The extra carrier's divergence constraint solves its third component polynomially; the apparent Schwarzschild-coordinate pole is a chart artifact.

Theorem 7.1 (Analytic ingoing curvature family)**.** *For axial $\ell = 2$ and $\omega \in \mathbb{R} \setminus \{0\}$, the future horizon $r = 2m$ is a regular singular point of the extra first-order radial system. Its residue spectrum is*

$$\{0, 0, -4im\omega, -2 - 4im\omega\}, \tag{21}$$

and the zero eigenspace has dimension two. Hence there is a two-parameter family of ψ-solutions analytic at the future horizon, with no leading logarithm at the zero exponent.

The restriction excludes $\omega = 0$, where the algebraic multiplicity of the zero residue exponent changes, and makes no claim at resonant complex frequencies. Most importantly, the theorem counts the curvature carrier, not a quotient of metric solutions. The currently certified dimensions are:

Space	Ingoing/analytic dimension	Status
Einstein Regge–Wheeler carrier	1	second-order control
Ricci carrier ψ	2	exact residue calculation
Full Bach metric solution space	not certified	reconstruction/domain open
Metric quotient by the Einstein kernel	not certified	lift and gauge open

Thus future-horizon regularity does not force $\psi = 0$. Explicit metric lifts are constructed in the flux fixtures below, but the exact sequence (20) has not yet been split globally.

8 Action-derived horizon flux

Substituting two on-shell Regge–Wheeler solutions into the current (3) gives

$$F^r_{EE} = -\frac{192\pi\alpha(\omega_1^2 - \omega_2^2)}{5\omega_1\omega_2 r}\Psi_1\Psi_2. \tag{22}$$

For conjugate pairs $\omega_2 = \pm\omega_1$, this vanishes identically.

Theorem 8.1 (Exact Einstein isotropy)**.** *On Schwarzschild in axial $\ell = 2$, the pure-Weyl Lee–Wald pairing of an Einstein Regge–Wheeler mode with its conjugate vanishes exactly:*

$$F^r(E_\omega, \overline{E_\omega}) = 0. \tag{23}$$

Thus the Einstein line is isotropic in the presymplectic structure inherited from the pure C^2 action. This statement does not identify the Einstein symplectic form with the ordinary Einstein–Hilbert one.

Proposition 8.2 (Controlled horizon-pairing fixtures)**.** *For $m = 1$, axial $\ell = 2$, and conjugate-frequency Hermitian pairings, order-16 ingoing series give*

$$F^r(E_\omega, \overline{X_\omega}) \neq 0, \qquad F^r(X_\omega, \overline{X_\omega}) \neq 0 \tag{24}$$

at producer frequencies $\omega = 3/5, 2/7$; an independent verifier repeats the nonvanishing gates at $\omega = 1/2$. At the two stored frequencies the values, divided by $\pi\alpha$ and sampled at two interior radii, are approximately

ω	$F^r(X, \bar{X})/(\pi\alpha)$	$F^r(E, \bar{X})/(\pi\alpha)$
3/5	$-330.58i, -329.98i$	$-218.17 + 43.55i, -215.97 + 51.53i$
2/7	$-71.92i, -71.95i$	$-162.82 + 27.99i, -163.92 + 33.03i$

The exact Regge–Wheeler null identity is the in-run control. These data are controlled numerical fixtures: they prove neither symbolic nonvanishing for all real ω nor an interval-certified error bound.

For a chosen lift basis (E, X), the horizon block has the form

$$\Omega_h = \begin{pmatrix} 0 & a \\ -\bar{a} & b \end{pmatrix}. \tag{25}$$

Under the admissible lift change $X \mapsto X + cE$, the cross entry a is invariant while b changes. The robust fixture-level conclusions are therefore Einstein isotropy, nonzero cross pairing, and full rank of the tested two-dimensional block. The sign of iF^r_{XX} for one chosen lift is basis dependent and is not promoted to a branch signature or Hilbert norm.

The stored report also records a repaired calculation: an earlier flux evaluation dropped the Fourier factor $e^{i\omega t}$, causing all time derivatives to vanish silently. The exact Regge–Wheeler null control exposed the defect; all displayed fixtures use the repaired full-time current.

9 Asymptotic symbol and endpoint nonselection

At real frequency and large radius, the extra carrier has dispersion

$$(\lambda_r^2 - \omega^2)^2 = 0 \tag{26}$$

and growth exponents

$$\sigma \in \{\pm 2im\omega,\ \pm 2im\omega - 1\}. \tag{27}$$

The simple-root reading would identify Coulomb logarithmic phases with amplitudes of order r^0 and r^{-1}. But the characteristic is *double*. The polynomial alone does not determine geometric multiplicity. As in $(D^2+\omega^2)^2 h = 0$, a Jordan chain may produce $r_* e^{\pm i\omega r_*}$ even though every characteristic exponent is oscillatory. The axial Ricci-carrier recurrences are known to all formal orders for every integer $\ell \geq 2$ and real $\omega \neq 0$. The earlier generic-ℓ metric lifts, however, did not impose the independent $v\varphi$ Ricci row. The complete metric endpoint module is certified here only at $\ell = 2$ on the real pilot interval. In the polar calculation below we use only the maximal restriction-stable module through the certified depth. Two additional terminal-only prefix vectors still require their next compatibility test and are not promoted to formal solutions.

Theorem 9.1 (Endpoint regularity-and-leading-symbol nonselection)**.** *On the declared Schwarzschild axial $\ell = 2$, nonzero-real-frequency mode carrier, future-horizon analyticity does not imply $\delta R_{ab} = 0$. Moreover, the presently certified leading characteristic polynomial and formal exponents do not supply an outer condition that distinguishes the Ricci carrier from the Einstein characteristic. Hence these tested local endpoint diagnostics do not select the Einstein kernel.*

This deliberately does *not* say that every ordinary asymptotically flat radiation condition admits an additional metric mode. That stronger statement requires convergence or another completion of the formal series, falloff in a regular tetrad or asymptotic gauge, a differentiable Lee–Wald phase space, and a horizon-to-infinity connection map. None is supplied by the formal calculations below.

9.1 Complete axial reconstruction on the $\ell = 2$ pilot

The earlier generic-angular metric audit imposed the $x\varphi$ and $r\varphi$ reconstruction equations but omitted the independent $v\varphi$ equation. Put

$$C = \frac{\delta R_{v\varphi}}{S_2} - P. \tag{28}$$

On every solution of the two differential rows and the Ricci-carrier equations, exact differentiation gives

$$C' = -\frac{2}{r}C, \qquad \kappa = r^2 C = \text{constant}. \tag{29}$$

The complete reconstruction is therefore the $\kappa = 0$ fibre. Solving this algebraic constraint for H_0 produces an exact six-state flow on $(P, P', Q, Q', H_1, H_1')$ and the short exact sequence

$$0 \longrightarrow \mathcal{E}^2_{\text{ker}} \longrightarrow \mathcal{E}^6_{\text{Bach,ax}} \longrightarrow \mathcal{E}^4_{\text{Ricci}} \longrightarrow 0. \tag{30}$$

It supplies six independent formal columns at each endpoint: four carrier lifts and two complete Einstein-kernel columns. This statement is exact for $\ell = 2$, $M = 1$, and real $\omega \in [1/2, 3/4]$; it is not a generic-ℓ metric theorem.

The legacy Phase-2 X_0 has

$$C_{X_0} = \frac{i(\omega - 18i)}{2\omega^2} r^{-2}. \tag{31}$$

It becomes a complete non-Einstein lift after an exact transverse addition. The vector formerly labelled E_0 is one half of that transverse vector and is not itself a complete Einstein solution; all claims using it as an Einstein representative are superseded.

For the repaired X_0, the literal current audit has no coefficient at any nonintegrable power $p \geq -1$. Its first self coefficient is

$$F^v(X_0, \overline{X}_0) = \frac{32 i\pi\alpha_{\mathrm{W}}(540 - \omega^2)}{15\omega^3(\omega^2 + 4)} r^{-2} + O(r^{-3}), \tag{32}$$

which is nonzero on the pilot interval. The same finite class is preserved under the true rate-zero Einstein-kernel shear. It is *not* preserved under an arbitrary Einstein shear: the true oscillatory Einstein kernel has cross term

$$F^v(E_{\mathrm{osc}}, \overline{X}_0) = \frac{48\pi\alpha_{\mathrm{W}}\omega^3(4\omega + i)}{5} e^{-2i\omega r} r^{3-4i\omega} + \cdots, \tag{33}$$

with a nonzero coefficient throughout the real pilot interval.

Theorem 9.2 (Scoped axial reconstruction and radial disposition)**.** *For $\ell = 2$, $M = 1$, and real $\omega \in [1/2, 3/4]$, the complete axial formal endpoint module has dimension six. Its repaired X_0 direction is non-Einstein and radially finite in the fixed representative and under true rate-zero Einstein shears, while true oscillatory Einstein shears produce divergent cross terms. Thus the pilot supplies a finite formal non-Einstein direction but no representative-independent axial selection or quotient theorem.*

This is an exact formal asymptotic coefficient audit. It establishes no convergence, finite-flux phase space, horizon-to-infinity matching, scattering channel, stability, or CPT property. The other axial additional direction and the generic angular tower require separate all-row audits.

9.2 Polar mixed finite line and its exceptional wall

On the maximal restriction-stable polar module, choose the ordered formal basis (E, X_0, X_1, X_2). The denominator-cleared current matrices at powers $p = 0, -1, -2$ are anti-Hermitian. The $p = 0$ matrix has rank three and a one-dimensional mixed Einstein/additional radical. If y spans that line, then

$$y^\dagger N^{(-1)} y = 0, \tag{34}$$

while its first finite coefficient factors as

$$y^\dagger N^{(-2)} y = C\,\omega^{51}\Lambda^3(\Lambda - 2)^5 P_6^2 P_{20} Q_{21}(\Lambda, \omega^2). \tag{35}$$

The constant C is nonzero, and the exact sum-of-squares and resultant certificates show that $P_6 P_{20} \neq 0$ for integer $\ell \geq 2$ and real $\omega \neq 0$. Therefore the filtered line is generically nonradical. At $Q_{21} = 0$ its deeper disposition remains open.

Writing $x = \widehat{\omega}^2 > 0$, the exact positive-root phase diagram is

Harmonic	positive x-roots	real nonzero $\widehat{\omega}$-roots
$\ell = 2$	0	0
$\ell = 3$	3	6
$4 \leq \ell \leq 10$	1	2
$11 \leq \ell \leq 40$	3	6
$\ell \geq 41$	1	2

At the legacy fixture $(\Lambda, x) = (6, 9/25)$,

$$Q_{21}(6, 9/25) = -\frac{174226120816040380076641138108451235935620694016}{227373675443232059478759765625} \neq 0. \tag{36}$$

The previously recorded longer rational was $Q_{21}(6, 81/625)$, caused by squaring x twice; it was neither the fixture value nor a Hilbert norm.

Theorem 9.3 (Generic polar formal radial disposition)**.** *Away from the explicit Q_{21} wall, a one-dimensional mixed Einstein/additional polar line is finite through the nonintegrable layers and has nonzero first finite pairing. Thus formal radial pairing finiteness does not canonically select a polar Einstein representative.*

This theorem neither extends the terminal-only polar prefixes nor matches the mixed line to future-horizon-regular data. In particular, the axial and polar infinity results do not identify a global solution admitted at both ends of the exterior.

9.3 Local Cauchy selection remains available

There is no no-go against local causal selection. On the globally hyperbolic exterior with Cauchy surface $\Sigma = \{t = \text{const}, r > 2m\}$, the local Cauchy condition

$$\psi|_\Sigma = 0, \qquad \nabla_n \psi|_\Sigma = 0 \tag{37}$$

propagates $\psi = 0$ in the axial sector unconditionally: the carrier operator equals $\frac{1}{2}\Box + C \circ (-)$ exactly there (the trace vanishes identically on the class), is normally hyperbolic, and its Bianchi constraint obeys the exact transport identity $\nabla^a (L\psi)_{a\varphi} = \frac{1}{2}\Box B_\varphi$. In the polar sector the operator is identically traceless at the PDE level, the conformal direction is uncontrolled, and $\psi_{\text{conf}}(t^4\chi(r)P_2)$ is an exact nonuniqueness witness with vanishing Cauchy data; the certified conformal trace relation $S(\psi_{\text{conf}}(\Phi)) = -3\Box\Phi$ then reduces any zero-data solution to the conformal-gauge orbit, so local Cauchy data selects the Einstein image exactly *modulo conformal gauge* (and exactly on the traceless slice). Dropping the normal-derivative datum is certified to fail by a time-odd mode witness. The open physical questions are whether this restriction is preserved by nonlinear interactions and whether it follows from a differentiable asymptotic phase space rather than being imposed by hand.

10 Polar chain: composition, reach, current, and disposition

For polar $\ell = 2$ Regge–Wheeler-gauge perturbations, the trace variation δR need not vanish. The axial formula acquires two universal trace terms.

Corollary 10.1 (Polar trace-coupled composition)**.** *On the certified polar $\ell = 2$ carrier, Proposition 6.1 gives*

$$\boxed{\delta B_{ab} = \frac{1}{2}\Box(\delta R_{ab}) + C_{acbd}(\delta R)^{cd} - \frac{1}{6}\nabla_a\nabla_b\delta R - \frac{1}{12}g_{ab}\Box\delta R.} \tag{38}$$

The Einstein polar kernel injects by $\delta R_{ab} = 0$, while the curvature image obeys a second-order trace-coupled Lichnerowicz system for $(\delta R_{ab}, \delta R)$.

The polar chain is now certified through the same stations as the axial chain. Four structural facts organise it.

(i) Polar Einstein kernel. In the t-chart Regge–Wheeler polar gauge the traceless angular row of δR_{ab} equals $(H_0 - H_2)/2$, forcing $H_2 = H_0$; the remaining rows reduce exactly to a two-dimensional

first-order system $dY/dr = M(r)Y$, $Y = (K, H_1)$, with H_0 algebraic. In horizon-adapted variables (K, BH_1) the ingoing residue spectrum is $\{0, -4im\omega\}$, matching the axial Regge–Wheeler control. A Schrödinger-form master scalar with ω-free potential was *not* found within the searched bounded rational ansatz classes; that anchor is recorded fail-closed open and nothing below depends on it.

(ii) Conformal gauge and polar reach. The polar carrier system on the Bianchi-constrained $(\delta R_{ab}, \delta R)$ is exactly three second-order equations in four radial functions: the operator (38) is identically traceless and divergence-free on the constraint surface, and the one-function underdeterminacy is linearized conformal gauge ($h = \Phi g$ is pure gauge and $\psi_{\mathrm{conf}} = \delta\mathrm{Ric}[\Phi g]$ annihilates every row for arbitrary Φ); this direction is absent in the axial sector. On the algebraic traceless slice the principal part diagonalises and the future horizon is a regular singular point with residue spectrum $\{0, 0, 0,\ 1 - 4im\omega,\ -1 - 4im\omega,\ -3 - 4im\omega\}$. Quotienting the regular conformal-gauge direction leaves a *two*-parameter physical ingoing-analytic polar carrier family at every $\omega \in \mathbb{R} \setminus \{0\}$ — the polar twin of the axial reach theorem.

(iii) Polar current. The sphere-integrated action-derived polar bilinear obeys the same off-shell 4α identity. On shell of the two-dimensional polar Einstein system every bilinear coefficient carries the factor $(\omega_1 + \omega_2)$: the polar Einstein self-block is exactly null for conjugate pairs, as in the axial sector. In addition the conformal direction Φg pairs to zero with an *arbitrary* off-shell polar perturbation: the sphere-integrated polar presymplectic form needs no conformal quotient at the bilinear level. Composition fixtures ($\omega = 3/5$, two interior radii, exact arithmetic) certify that the realized Ricci image closes on the full three-dimensional analytic carrier space — every ingoing-analytic polar carrier mode lifts to a metric perturbation with all seven δRic rows verified — and that the Einstein–carrier cross pairing is nonzero. Cross-pairing nonvanishing is representative-independent because the Einstein block is null; the positive carrier self-norms are certified at the canonical composed representatives only, and the representative-invariant sign theory is open.

(iv) Polar disposition. The polar carrier's asymptotic symbol is $(\lambda^2 - \omega^2)^3$: it propagates on the Einstein characteristics, with Coulomb log-phases and amplitude falloffs r^{-1}, r^{-2}, r^{-3} — all decaying at real frequencies, strictly stronger than the axial r^0, r^{-1} case. The tested endpoint diagnostics (future-horizon analyticity, leading outer symbol and falloff class) therefore fail to separate the polar Ricci carrier from the polar Einstein carrier, exactly as in the axial sector: the endpoint-nonselection result is now *parity complete* at the $\ell = 2$ linear mode level. As before, this is not a classification of all local differential boundary operators.

(v) Polar formal infinity module. A later full-seven-row audit found that the shallow four-state master used in the earlier norm-selection table contained a source-zero direction that did not satisfy the complete Ricci system. The restriction-stable replacement and its current filtration are summarized by Equation (35): a mixed Einstein/additional line is finite and generically nonradical. The former power-enhanced/logarithmic fixture and parity-complete Einstein-only norm-selection theorem are therefore superseded. The independently verified polar composition, horizon reach, Einstein isotropy, conformal degeneracy, and controlled horizon pairings in items (i)–(iv) are unaffected.

11 Horizon monodromy temperature

One statement of Hawking type is available at the REDUCED-MODE level without any Lorentzian quantum construction. The surface gravity of the certified background is $\kappa = 1/(4m)$ with $T_H = \kappa/2\pi = 1/(8\pi m)$; the geometric-clock value agrees exactly with the certified normalized-frame first-law temperature.

Theorem 11.1 (Monodromy universality)**.** *Under the horizon continuation $\rho \to e^{2\pi i}\rho$, every* *certified ingoing residue exponent of the four $\ell = 2$ mode families (axial and polar, Einstein and Ricci carrier) has continuation factor exactly* 1 *or* $e^{8\pi m\omega} = e^{\omega/T_H}$*, and every family contains exponents with the nontrivial factor.*

The integer parts of the exponents are monodromy-trivial; the universal $-4im\omega$ part carries the thermal factor. Hence the Boltzmann ratio $|\beta/\alpha|^2 = e^{-\omega/T_H}$ of the Damour–Ruffini continuation is identical for the Einstein and additional curvature families in both parity sectors: at the mode level, the additional carrier is thermally weighted at exactly the Hawking temperature, and by the certified nonzero mixed pairings it participates in the horizon flux with that weight. This is a mode-level (REDUCED-MODE) statement only: no Hadamard state, renormalized stress tensor, grey-body factor, luminosity, back-reaction, or LORENTZIAN-CAUSAL Hawking theorem is claimed, and none exists in the repository.

12 Result ledger

Question	Certified answer	Scope
Static Bach-flat black holes?	exact Laurent-family classification and three-horizon fixture; Einstein sheet corrected	local/static
Differentiable energy?	residual-basic normalized generator, exact static first law	$u \neq 0$; signed orientation; $f(J)$ freedom
Einstein axial waves present?	yes; exact Regge–Wheeler injection	Schwarzschild, axial $\ell = 2$
Additional axial content?	nonzero second-order Ricci carrier; no canonical metric split	same
Reach future horizon?	two analytic ingoing ψ amplitudes	nonzero-real-frequency carrier
Carry current?	Einstein self-block exactly null; mixed/additional blocks nonzero in fixtures	exact identity plus three frequencies
Selected by tested endpoints?	no: horizon and leading-symbol tests do not force $\psi = 0$	not a general causal no-go
Formal radial infinity?	On the complete axial $\ell = 2$ pilot, repaired X_0 is finite in the fixed representative and under rate-zero Einstein shears, but oscillatory Einstein shears cross it divergently; the polar sector has a generic-angular mixed finite line	no representative-invariant axial quotient, phase space, norm, or global matching
Polar exceptional wall?	generically nonradical finite line; exact wall $Q_{21}(\ell(\ell+1), \widehat{\omega}^2) = 0$ with certified harmonic root counts	deeper filtration open on the wall
Einstein–extra horizon pairing invariant?	Einstein block isotropic; nonzero cross pairing and full rank in controlled fixtures; extra self-pairing sign not invariant	axial $\ell = 2$, real $\omega \neq 0$
Complex-frequency structure?	axial modes and cross current continue meromorphically with exact singular set $\{i, i/2\}$ (polar not activated); the Einstein connection problem is conditional on an uncomputed Wronskian	REDUCED-MODE; not a spectrum or BVP theorem
Static ($\omega = 0$) sectors?	log-classified: two-parameter log-free families both parities; polar Einstein variables degenerate	classification only; static composition open
Polar composition?	exact trace-coupled identity; realized image closes on the full analytic carrier space	Schwarzschild, polar $\ell = 2$
Polar reach and current?	two physical ingoing amplitudes modulo conformal gauge; Einstein self-block null; nonzero mixed fixtures	conformal direction an exact off-shell degeneracy
Polar disposition?	Einstein characteristics, all-decaying Coulomb asymptotics; endpoint nonselection parity complete	real frequencies, $\ell = 2$
Thermal weighting?	monodromy factor $e^{-\omega/T_H}$ universal across branches and parities	REDUCED-MODE; no quantum state
Stable/ringing?	not established	complex frequencies open; no QNM

13 What is not proved

The paper does not establish:

- a complete exterior initial-boundary well-posedness theorem: the complex-frequency Einstein connection is conditional on the nonvanishing of an uncomputed confluent-Heun Wronskian, while no outgoing function space has been constructed for the additional directions;
- convergence or summability of the generic-ℓ formal series, axial X_2, the deeper polar filtration on the $Q_{21} = 0$ wall, the terminal-only polar prefixes, or a non-Einstein-background Ricci-carrier composition;
- a canonical splitting of the Bach solution space, or complete metric reconstruction from every Ricci-carrier solution;
- a representative-invariant sign theory for the carrier self-pairing (the null-quotient pairing);
- mode stability, quasinormal modes, or ringdown waveforms, or the connection-Wronskian zeros that would define them: the certified complex-frequency statement is only that the axial modes and cross current continue *meromorphically* with exact singular set $\{i, i/2\}$ (polar not activated), which is not a spectrum;
- a second-order physical-process first law or nonlinear horizon dynamics;
- asymptotically flat conformal-gravity scattering with a complete boundary phase space, metric/tetrad falloffs, gauge-audited charge algebra, and horizon-to-infinity connection map: formal radial integrability is not a substitute for any of these;
- Hawking states, particles, grey-body factors, luminosity, a quantum BRST quotient, positivity, or unitarity: the monodromy temperature is a mode-level continuation factor, not a quantum state construction.

In particular, the paper withdraws the broad phrase "no local causal truncation." Zero Cauchy data for the Ricci carrier is a local linear truncation. What is established is failure of the tested endpoint diagnostics to imply that truncation.

14 Certificate and reproduction ledger

The manuscript is an OUTLINE-ALLOWED/DRAFT-ALLOWED integration of committed certificates at repository baseline recorded in its generated claim map.

Claim	Authoritative certificate
Static spherical family	`black_hole_programme/certificates/BH0_STATIC_SPHERICAL_BACKGROUND.json`
Normalized generator and first law	`black_hole_programme/certificates/BH1A_NORMALIZED_GENERATOR.json`
Linear spherical dynamical extension	`black_hole_programme/certificates/BH1B_DYNAMICAL_EXTENSION.json`
Axial operator and Ricci–Bach composition	`black_hole_programme/certificates/BH2A_AXIAL_OPERATOR.json`
Extra horizon reach	`black_hole_programme/certificates/BH2A_HORIZON_REACH.json`
Action-derived axial current	`black_hole_programme/certificates/BH2A_FLUX_MATRIX.json`
Mixed and extra horizon flux fixtures	`black_hole_programme/certificates/BH2A_CROSS_FLUX.json`
Axial causal disposition	`black_hole_programme/certificates/BH2A_CAUSAL_DISPOSITION.json`
Polar trace-coupled split	`black_hole_programme/certificates/BH2B_POLAR_SPLIT.json`
Polar carrier horizon reach (conformal gauge quotient)	`black_hole_programme/certificates/BH2B_POLAR_REACH.json`
Polar Einstein two-dimensional reduction	`black_hole_programme/certificates/BH2B_POLAR_EINSTEIN.json`
Polar Einstein self-block null; conformal degeneracy	`black_hole_programme/certificates/BH2B_POLAR_FLUX.json`
Polar composition and mixed flux fixtures	`black_hole_programme/certificates/BH2B_POLAR_CROSS_FLUX.json`
Polar causal disposition	`black_hole_programme/certificates/BH2B_POLAR_DISPOSITION.json`
Horizon monodromy temperature	`black_hole_programme/certificates/BH4_HAWKING_MONODROMY.json`
Static-sector log classification	`black_hole_programme/certificates/BH2_OMEGA_ZERO.json`
Generic-ℓ axial recurrences and Einstein current	`black_hole_programme/phase2/general_l_axial_asymptotics/certificate.json`; `black_hole_programme/phase2/general_l_axial_current/certificate.json`
Complete axial six-dimensional module, repaired X_0, and legacy E_0 supersession	`black_hole_programme/phase3/axial_complete_reconstruction_repair/certificate.json`; `reports/phase3-black-hole-axial-complete-reconstruction-repair-2026-07-22.md`
Polar restriction-stable module and invariant current filtration	`black_hole_programme/phase2/general_l_polar_extendible_current_closure/certificate.json`
Generic-ℓ two-parity formal-infinity phase diagram	`black_hole_programme/phase2/generic_l_synthesis/certificate.json`
Local Cauchy truncation selects Einstein modulo conformal gauge	`black_hole_programme/certificates/BH_LOCAL_EINSTEIN_CAUCHY_TRUNCATION.json`
Horizon cross invariant and complex-frequency continuation	`black_hole_programme/certificates/BH2_SYMBOLIC_CROSS_INVARIANT.json`; `black_hole_programme/certificates/BH3_ANALYTIC_CONTINUATION_GATE.json`

Every certificate records a schema, exact or controlled-numerical payload, independent verifier, mutations or controls, command receipt, dependency tags, and missing claims. The former axial and polar infinity-flux certificates, the infinity-selection half of the endpoint assembly, and the additional-log-tail half of the exterior boundary-value gate are historical inputs, not active authorities for infinity selection. Their unaffected Einstein, horizon, and leading-symbol components are represented above by independent certificates. The exact superseded identifiers are recorded in the machine-readable claim map. A theorem about asymptotically flat radiation still requires the regular-tetrad falloff, differentiable phase-space, charge-algebra, and global connection constructions.

15 Outlook

The formal infinity problem now has a different shape from the one suggested by the original fixture. The complete axial $\ell = 2$ pilot contains a finite repaired non-Einstein direction, but that finite class is stable only under the rate-zero Einstein shear and not under the true oscillatory shear. The polar module separately contains a generic-angular, generically nonradical mixed finite line, with a finite algebraic set of exceptional frequencies at each harmonic. These facts neither establish a representative-independent axial quotient nor restore an Einstein-selection principle.

The next decisive map is global:

$$\text{future-horizon data} \longrightarrow \text{exact exterior connection} \longrightarrow \text{differentiable asymptotic phase space.}$$

It must include a regular tetrad or gauge, convergence or another completion of the formal series, a gauge-audited Lee–Wald charge algebra, and the connection matrix between the two ends. The present exact rational and polynomial data neither prove nor obstruct that map. Complex-frequency work is likewise limited to the existing meromorphic axial continuation; no Wronskian zeros, quasinormal spectrum, or stability theorem have been computed.

The present result already fixes an important baseline:

Additional curvature directions occur in the future-horizon carrier, and independently the formal infinity modules contain finite non-Einstein or mixed directions. Neither leading endpoint diagnostics nor formal radial pairing finiteness selects the Einstein kernel. Whether one global solution realizes both endpoint properties remains open.

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